

STEFANSSON ISLAND, NU, Canada

WMO: 710170

Lat: 73.766N

Lon: 105.296W

Elev: 11

StdP: 101.19

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-42.0	-40.6	-46.5	0.0	-41.9	-44.9	0.0	-40.6	14.1	-24.2	13.0	-26.6	3.7	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	12.2	9.8	9.1	7.3	6.3	5.1	9.9	12.2	7.5	9.0	5.2	6.3	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.7	12.2	5.9	5.8	9.0	4.0	5.0	6.1	29.3	12.2	23.7	9.1	19.0	6.4	16.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.3	9.9	DB	-43.5	16.9	1.4	2.1	-44.5	18.4	-45.3	19.7	-46.1	20.9	-47.2	22.4	(4)
(5)			WB	-43.4	12.9	1.2	1.2	-44.3	13.8	-45.0	14.5	-45.7	15.2	-46.6	16.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-15.4	-30.3	-31.7	-30.9	-22.9	-9.7	0.2	3.1	1.8	-3.4	-12.0	-21.4	-28.4
	DBStd	13.93	5.51	4.93	6.05	4.77	5.38	2.66	2.73	2.96	3.33	5.45	5.53	5.69
	HDD10.0	9265	1250	1168	1268	986	612	294	214	254	403	682	943	1189
	HDD18.3	12303	1508	1402	1527	1236	871	544	471	511	653	940	1193	1447
	CDD10.0	3	0	0	0	0	0	0	2	2	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.6	4.8	4.8	5.2	4.8	4.4	4.2	4.2	4.4	5.0	4.6	4.2	4.4
Precipitation	PrecAvg	145	4	3	5	6	10	16	26	28	20	13	8	5
	PrecMax	233	10	12	13	19	30	42	52	79	54	34	15	12
	PrecMin	53	0	0	2	1	1	2	4	7	4	3	1	1
	PrecStd	33	3	3	3	5	6	10	11	17	10	8	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-14.0	-15.1	-11.1	-7.5	1.8	11.1	16.6	16.2	5.6	-0.6	-5.5	-6.8
		MCWB	-14.3	-15.3	-11.3	-7.7	0.4	8.7	12.0	12.5	4.5	-1.0	-5.8	-6.9
	2%	DB	-17.2	-19.0	-14.4	-10.8	0.3	7.6	13.2	12.0	3.4	-1.9	-8.8	-14.5
		MCWB	-17.7	-19.2	-14.7	-11.3	-0.6	6.1	10.6	9.8	2.7	-2.2	-9.1	-14.8
	5%	DB	-19.5	-21.0	-18.2	-13.2	-0.7	5.0	10.1	8.6	1.6	-3.1	-11.2	-16.9
		MCWB	-19.8	-21.2	-18.4	-13.6	-1.5	3.9	8.2	7.0	1.0	-3.5	-11.5	-17.2
	10%	DB	-22.2	-23.3	-21.6	-15.3	-2.1	3.1	7.1	5.7	0.4	-4.5	-13.7	-19.8
		MCWB	-22.4	-23.5	-21.8	-15.8	-2.9	2.2	5.6	4.7	-0.2	-4.9	-14.0	-20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-14.4	-15.3	-11.4	-7.7	0.6	8.7	12.2	12.9	4.7	-0.8	-5.8	-7.0
		MCDB	-14.1	-15.0	-11.2	-7.4	1.5	11.0	15.8	16.2	5.5	-0.6	-5.6	-6.8
	2%	WB	-17.5	-19.3	-14.7	-11.3	-0.4	6.0	10.6	9.8	2.7	-2.2	-9.0	-14.7
		MCDB	-17.2	-19.1	-14.5	-10.9	0.3	7.6	13.3	11.9	3.3	-1.9	-8.8	-14.5
	5%	WB	-19.8	-21.2	-18.4	-13.5	-1.4	3.9	8.3	7.0	1.1	-3.5	-11.5	-17.1
		MCDB	-19.5	-21.1	-18.1	-13.1	-0.7	4.9	10.0	8.5	1.5	-3.2	-11.3	-16.9
	10%	WB	-22.4	-23.5	-21.7	-15.7	-2.9	2.2	5.7	4.7	-0.1	-4.9	-14.0	-20.0
		MCDB	-22.2	-23.3	-21.5	-15.3	-2.2	3.0	6.9	5.7	0.3	-4.6	-13.8	-19.8
Mean Daily Temperature Range	5% DB	MDBR	6.2	6.8	7.2	8.2	6.4	3.9	4.6	4.2	3.2	5.6	6.6	6.6
		MCDBR	8.5	9.4	9.4	7.7	6.3	7.0	10.2	9.5	4.8	5.3	8.1	8.3
		MCWBR	8.3	9.3	9.3	7.4	5.6	5.8	8.2	7.8	4.3	5.3	8.2	8.3
	5% WB	MCDBR	8.6	9.3	9.3	7.6	6.0	6.8	10.1	9.2	4.7	5.3	8.1	8.3
		MCWBR	8.4	9.2	9.2	7.3	5.4	5.7	8.2	7.7	4.3	5.2	8.2	8.2
Clear-Sky Solar Irradiance	taub	N/A	0.162	0.232	0.284	0.309	0.346	0.328	0.319	0.270	0.181	N/A	N/A	
	taud	N/A	1.842	2.107	2.048	2.056	2.085	2.309	2.366	2.345	1.876	N/A	N/A	
	Ebn at Noon	0	373	717	803	837	812	824	776	685	295	0	0	
	Edh at Noon	0	25	64	105	123	125	96	79	53	21	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.13	1.36	4.03	6.00	6.59	5.32	3.05	1.33	0.31	0.00	0.00	
	RadStd	0.00	0.01	0.08	0.13	0.36	0.47	0.40	0.26	0.14	0.03	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page